MINI PROJECT (**CS41** Group 4)

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DISTANCE BETWEEN TWO CITIES

➢ Problem Statement 1

(Route Selection and Travel Feasibility Checker)

To design and implement a shell script based travel route planner that determines whether a user can reach a destination city based on the available fuel and cash.
The program should read multiple routes from a file, display all possible routes between a given source and destination, and allow the user to select a route. Based on the selected route distance, the script should analyze whether the journey can be completed directly with available fuel, by stopping to refill fuel using available cash, or if the journey is not possible due to insufficient fuel and cash.
The program should also identify and display the nearest route among all available routes.

Objective:-

- To design and implement a shell script based travel route planner.
- To determine travel feasibility based on available fuel and cash
- To read and display multiple travel routes from a file.
- To analyze whether the destination can be reached directly, after refueling,or not at all.
- To identify and display the nearest available route between the source and destination.

```bash
#!/bin/bash
# Specify the Bash shell for script execution

# ----------------- Constant Values -----------------

# Average distance covered per litre of fuel
KM_PER_LITRE=15

# Cost of fuel per litre
FUEL_PRICE=100

# File containing route details
DB="P1routes.txt"

# ----------------- User Input -----------------

# Read source city from user
echo "Enter Source City:"
read source

# Read destination city from user
echo "Enter Destination City:"
read destination

# Read available fuel in litres
echo "Enter fuel available (litres):"
read fuel

# Read available cash in rupees
echo "Enter cash available (₹):"
read cash
```

```bash
# ----------------- Distance Calculation -----------------

# Calculate distance possible using available fuel
fuel_dist=$((fuel * KM_PER_LITRE))

# Calculate additional distance possible by buying fuel using cash
cash_dist=$(((cash / FUEL_PRICE) * KM_PER_LITRE))

# Calculate total possible distance (fuel + cash)
total_possible=$((fuel_dist + cash_dist))

# Display distance analysis
echo
echo "Distance possible with current fuel = $fuel_dist km"
echo "Additional distance possible using cash = $cash_dist km"
echo "Total possible distance = $total_possible km"
echo "-----------------------------------------------"
```

```
# ----------------- Route Processing -----------------

# Initialize variables
found=0
min=99999
nearest=""
route_no=0

# Declare arrays to store route details
declare -a routes
declare -a distances

# Read route data from file
while read s d via dist
do
  # Check if route matches source and destination
  if [[ "$s" == "$source" && "$d" == "$destination" ]]; then
    found=1
    route_no=$((route_no + 1))

    # Store route details in arrays
    routes[$route_no]="$via"
    distances[$route_no]=$dist

    # Display available route
    echo "$route_no) $source -> $via -> $destination = $dist km"

    # Find the nearest route
    if [ $dist -lt $min ]; then
      min=$dist
      nearest=$via
    fi
  fi
done < "$DB"

# If no routes are found, exit the program
if [ $found -eq 0 ]; then
  echo "No routes found."
  exit
fi
```

```
# ----------------- Route Selection -----------------

# Ask user to choose a route
echo
echo "Enter route number you want to travel:"
read choice

# Get selected route details
chosen_dist=${distances[$choice]}
chosen_via=${routes[$choice]}

# Display selected route
echo
echo "You selected route via $chosen_via ($chosen_dist km)"
echo
echo "------ Travel Analysis ------"
```

```
# ------------------ Travel Feasibility Check ------------------

# Check if journey can be completed directly with available fuel
if [ $chosen_dist -le $fuel_dist ]; then
    echo "✅ You CAN reach the destination directly (fuel is sufficient)"

# Check if refueling is required but possible using cash
elif [ $chosen_dist -le $total_possible ]; then
    echo "⚠️Fuel insufficient, but cash is available"
    echo "🛑 Stop at $chosen_via to refill fuel"
    echo "➡️Then continue to $destination"

# Journey not possible due to insufficient fuel and cash
else
    echo "❌ Journey NOT possible (fuel + cash insufficient)"
fi

# ------------------ Nearest Route Display ------------------

# Display the nearest available route
echo
echo "Nearest route overall = $nearest ($min km)"
```

The shell script uses predefined constants such as distance per litre, fuel price, and route file name to simplify travel distance and cost calculations.

Fixed Values Used in the Shell Script

Variable Name	Fixed Value	Description
KM_PER_LITRE	15 km	Distance that can be travelled per litre of fuel
FUEL_PRICE	₹100	Cost of fuel per litre
DB	P1routes.txt	File containing available travel routes
min	99999 km	Initial value to find the shortest route
found	0	Flag to check whether routes are found
route_no	0	Counter to number the available routes

P1routes.txt File Description:-

The file P1routes.txt stores all available travel routes between different cities. Each line of the file contains four fields:

SourceCity DestinationCity ViaCity Distance

```
Pune Mumbai Lonavala 120
Pune Mumbai Khopoli 140
Pune Mumbai Karjat 145
Pune Mumbai Panvel 135
Delhi Amritsar Jalandhar 450
Delhi Amritsar Chandigarh 480
Manali Shimla Mandi 250
Manali Shimla Kullu 255
Pune Kokan Tamhini_Ghat 140
Pune Kokan Chiplun 300
Srinagar Leh Sonamarg 430
Srinagar Leh Pahalgam 480
Srinagar Leh Anantang 440
~
```

```
samruddhi-mate@samruddhi-mate-VirtualBox:~$ vim P1GRP4.sh
samruddhi-mate@samruddhi-mate-VirtualBox:~$ vim P1routes.txt
samruddhi-mate@samruddhi-mate-VirtualBox:~$ chmod +x P1GRP4.sh
samruddhi-mate@samruddhi-mate-VirtualBox:~$ bash P1GRP4.sh
Enter Source City:
Pune
Enter Destination City:
Mumbai
Enter fuel available (litres):
10
Enter cash available (₹):
100

Distance possible with current fuel = 150 km
Additional distance possible using cash = 15 km
Total possible distance = 165 km
-----------------------------------------------
1) Pune -> Lonavala -> Mumbai = 120 km
2) Pune -> Khopoli -> Mumbai = 140 km
3) Pune -> Karjat -> Mumbai = 145 km
4) Pune -> Panvel -> Mumbai = 135 km

Enter route number you want to travel:
4

You selected route via Panvel (135 km)

------ Travel Analysis ------
✅ You CAN reach the destination directly (fuel is sufficient)

Nearest route overall = Lonavala (120 km)
```

Conclusion:

The program effectively analyzes travel feasibility using fuel, cash, and route distance, showcasing practical use of shell scripting.

➢ Problem Statement 2

(Travel Planner Using Shell Script)

To design and implement a menu-driven shell script that acts as a Travel Planner.The program should calculate the travel cost from Pune to Jaipur based on the user-selected mode of transport and also compute the local sightseeing travel cost in Jaipur based on the selected tourist place and local transport mode.The script should interact with the user, accept inputs, perform arithmetic calculations, and display the final travel details and costs.

Objectives:-

- To understand the use of shell scripting for real-life applications
- To apply case statements for menu-based selection
- To perform arithmetic operations using shell script
- To accept user input and display formatted output

```
#!/bin/bash
# Shebang line to specify Bash shell

# Display welcome message
echo "-----------------------------------------------"
echo "Welcome to Travel Planner (Rajasthan Tour)"
echo "-----------------------------------------------"
echo

# ================= PART 1 : Pune to Jaipur =================
# This part calculates travel cost from Pune to Jaipur

# Distance from Pune to Jaipur in km
distance_pj=1200

# Rates per km for different travel modes
train_rate=2
flight_rate=6
bus_rate=3
car_rate=12

# Display journey details
echo "You are travelling from Pune to Jaipur (Rajasthan)"
echo "Distance : $distance_pj km"
echo
```

```
# Display travel mode options
echo "Choose mode of travel:"
echo "1. Train      (₹$train_rate per km)"
echo "2. Aeroplane  (₹$flight_rate per km)"
echo "3. Bus        (₹$bus_rate per km)"
echo "4. Car        (₹$car_rate per km)"
echo

# Read user choice for travel mode
read mode1
echo

# Select rate based on user choice
case $mode1 in
  1) rate=$train_rate; mode_name="Train" ;;
  2) rate=$flight_rate; mode_name="Aeroplane" ;;
  3) rate=$bus_rate; mode_name="Bus" ;;
  4) rate=$car_rate; mode_name="Car" ;;
  *) echo "Invalid choice"; exit ;;
esac

# Calculate total cost for Pune to Jaipur journey
cost1=$((distance_pj * rate))

# Display travel summary
echo "Mode Selected : $mode_name"
echo "Distance      : $distance_pj km"
echo "Cost          : ₹$cost1"
echo
echo "You have reached Jaipur successfully 😊"
echo
```

```
# ================= PART 2 : Local Sightseeing =================
# This part calculates cost for local sightseeing in Jaipur

# Display sightseeing options
echo "-----------------------------------------"
echo "Now choose a tourist place to visit in Jaipur"
echo "-----------------------------------------"
echo

echo "Tourist Places:"
echo "1. Amber Palace    (11 km)"
echo "2. Jal Mahal       (5 km)"
echo "3. Hawa Mahal      (4 km)"
echo "4. Jantar Mantar   (2 km)"
echo "5. City Palace     (3 km)"
echo "6. Nahargarh Fort  (6 km)"
echo

# Read tourist place choice
read place_no
echo

# Assign place name and distance based on choice
case $place_no in
  1) place="Amber Palace"; distance_local=11 ;;
  2) place="Jal Mahal"; distance_local=5 ;;
  3) place="Hawa Mahal"; distance_local=4 ;;
  4) place="Jantar Mantar"; distance_local=2 ;;
  5) place="City Palace"; distance_local=3 ;;
  6) place="Nahargarh Fort"; distance_local=6 ;;
  *) echo "Invalid place number"; exit ;;
esac
```

```
# Local travel rates per km
car_rate_l=12
bus_rate_l=5
auto_rate_l=8

# Display local travel options
echo "Choose local mode of travel:"
echo "1. Car   (₹$car_rate_l per km)"
echo "2. Bus   (₹$bus_rate_l per km)"
echo "3. Auto  (₹$auto_rate_l per km)"
echo

# Read local travel mode
read mode2
echo

# Select local travel rate
case $mode2 in
  1) rate_l=$car_rate_l; mode_l="Car" ;;
  2) rate_l=$bus_rate_l; mode_l="Bus" ;;
  3) rate_l=$auto_rate_l; mode_l="Auto" ;;
  *) echo "Invalid local mode"; exit ;;
esac

# Calculate local travel cost
cost2=$((distance_local * rate_l))

# Display final journey details
echo "-----------------------------------------"
echo "Next Destination : $place"
echo "Mode of Travel   : $mode_l"
echo "Distance         : $distance_local km"
echo "Cost             : ₹$cost2"
echo
echo "Have a happy journey 😊"
```

```
samruddhi-mate@samruddhi-mate-VirtualBox:~$ vim P2_GRP4.sh
samruddhi-mate@samruddhi-mate-VirtualBox:~$ chmod +x P2_GRP4.sh
samruddhi-mate@samruddhi-mate-VirtualBox:~$ bash P2_GRP4.sh
-----------------------------------------------
Welcome to Travel Planner (Rajasthan Tour)
-----------------------------------------------

You are travelling from Pune to Jaipur (Rajasthan)
Distance : 1200 km

Choose mode of travel:
1. Train      (₹2 per km)
2. Aeroplane  (₹6 per km)
3. Bus        (₹3 per km)
4. Car        (₹12 per km)

2

Mode Selected : Aeroplane
Distance      : 1200 km
Cost          : ₹7200

You have reached Jaipur successfully 😊

-----------------------------------------
Now choose a tourist place to visit in Jaipur
-----------------------------------------

Tourist Places:
1. Amber Palace    (11 km)
2. Jal Mahal       (5 km)
3. Hawa Mahal      (4 km)
4. Jantar Mantar   (2 km)
5. City Palace     (3 km)
6. Nahargarh Fort  (6 km)

1

Choose local mode of travel:
1. Car   (₹12 per km)
2. Bus   (₹5 per km)
3. Auto  (₹8 per km)

1

-----------------------------------------
Next Destination : Amber Palace
Mode of Travel   : Car
Distance         : 11 km
Cost             : ₹132

Have a happy journey 😊
```

Conclusion:-

This project emphasizes the role of distance between cities in calculating travel cost and selecting suitable transport modes using shell scripting.